



Computer Networks

Week # 06

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This Week Course Plan

- Transmission Modes
- Circuit switched and packet switched Networks
- Ethernet LAN Standard
- Overview of Wireless LAN

ACKNOWLEDGMENT

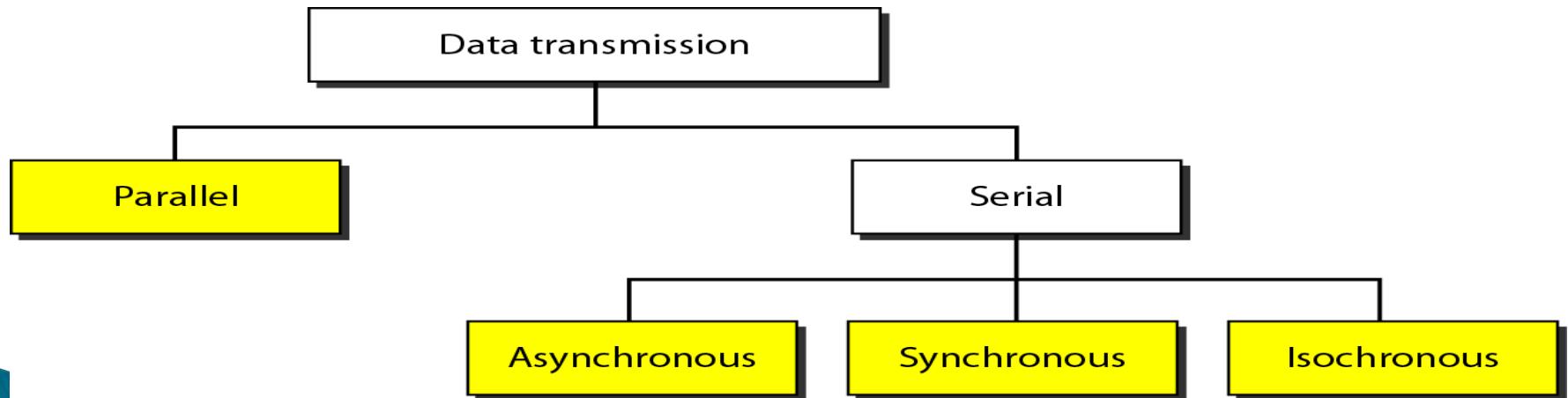
These lecture slides contain material from slides prepared by Behrouz Forouzan for his book Data Communication and Networking (4th/5th edition).

TRANSMISSION MODES

The transmission of binary data across a link can be accomplished in either parallel or serial mode

In **parallel mode**, multiple bits are sent with each clock tick.

In **serial mode**, 1 bit is sent with each clock tick.

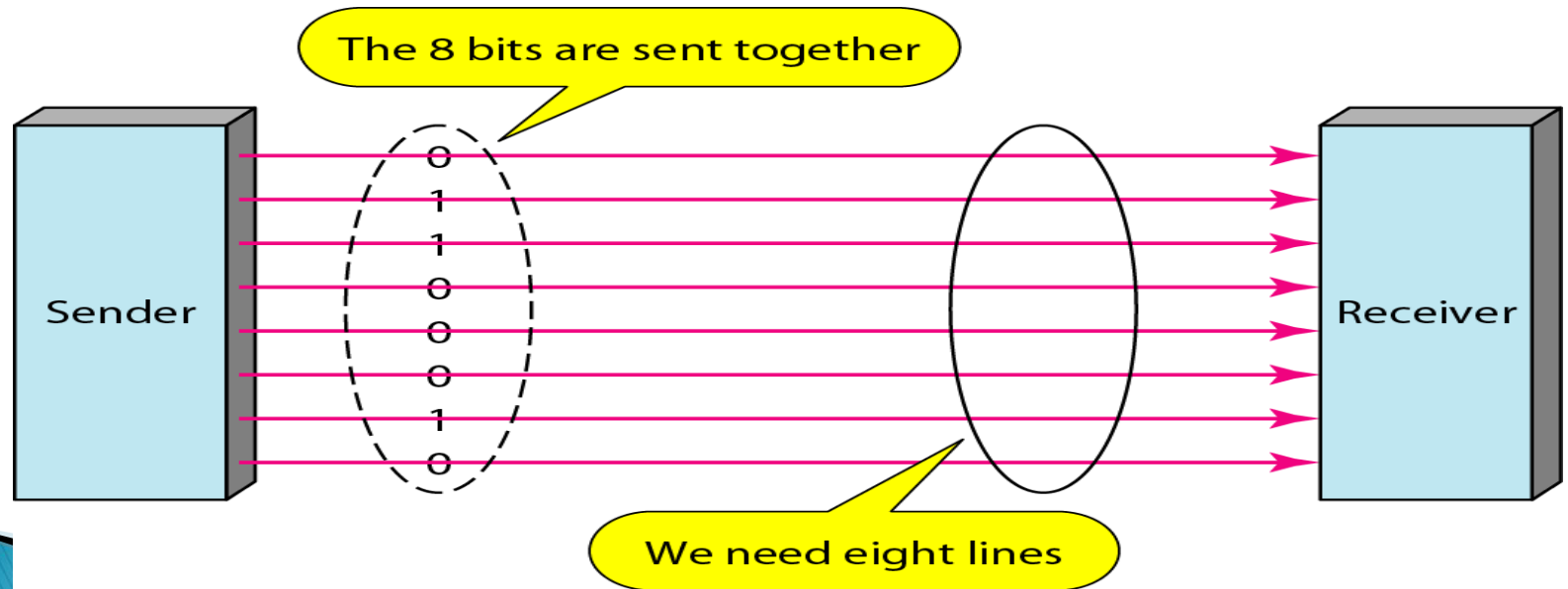


Parallel transmission

Use n wires to send n bits at a time; each bit has its own wire, so all n bits of one group can be transmitted with each clock tick between devices.

Advantage: speed (by a factor of n over serial)

Disadvantage: Cost



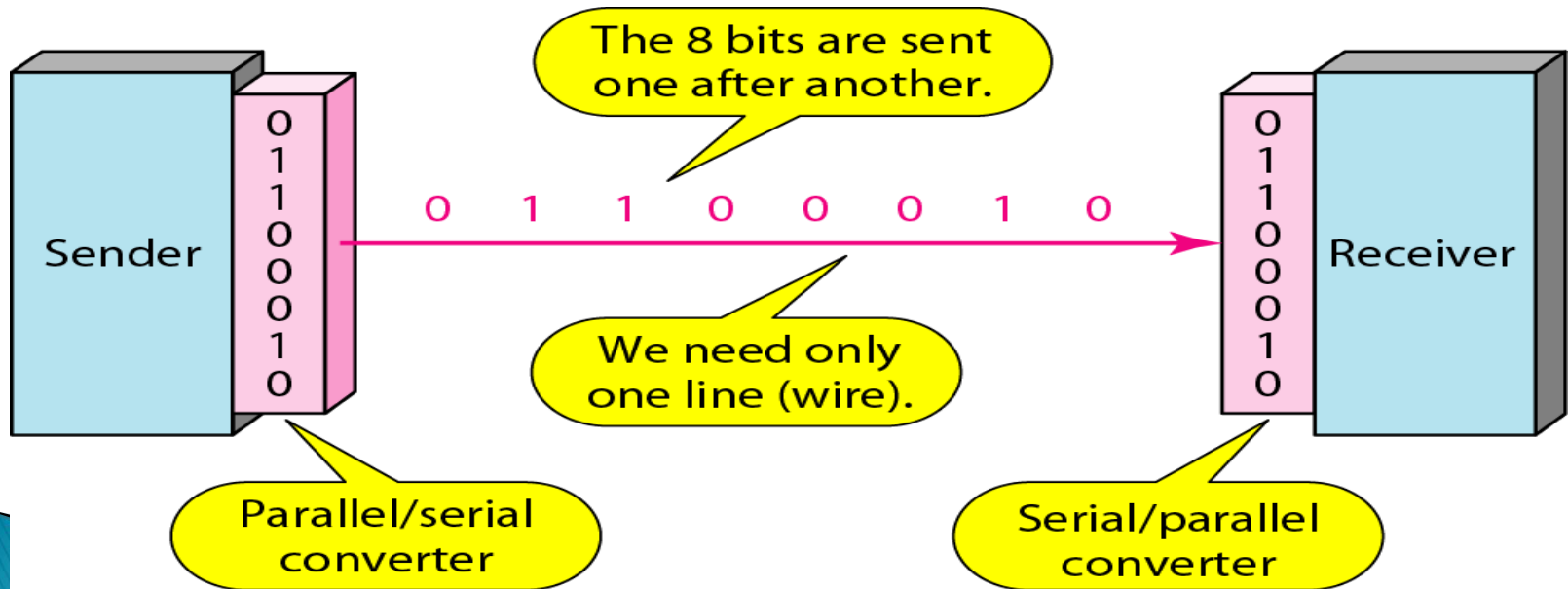
Serial transmission

One bit follows another which requires only one channel for communication between devices

Advantage: reduces cost (only one channel instead of n)

Overhead: Because communication within devices is parallel, we need conversion devices at the interface between sender and the line (parallel to serial) and between the line and the receiver (serial to parallel)

Serial transmission can be **asynchronous**, **synchronous** and **Isochronous**



Asynchronous transmission

In asynchronous transmission, signal timing is not important

- for synchronization

- we send ***1 start bit*** (0) to alert the receiver to the arrival of a new group, and

- and ***1 or more additional stop bits*** (1s) at the end of each byte to let the receiver to know that byte is finished

- There may be a gap between each byte

- Asynchronous here means “asynchronous at the byte level,

- but the bits are still synchronized;

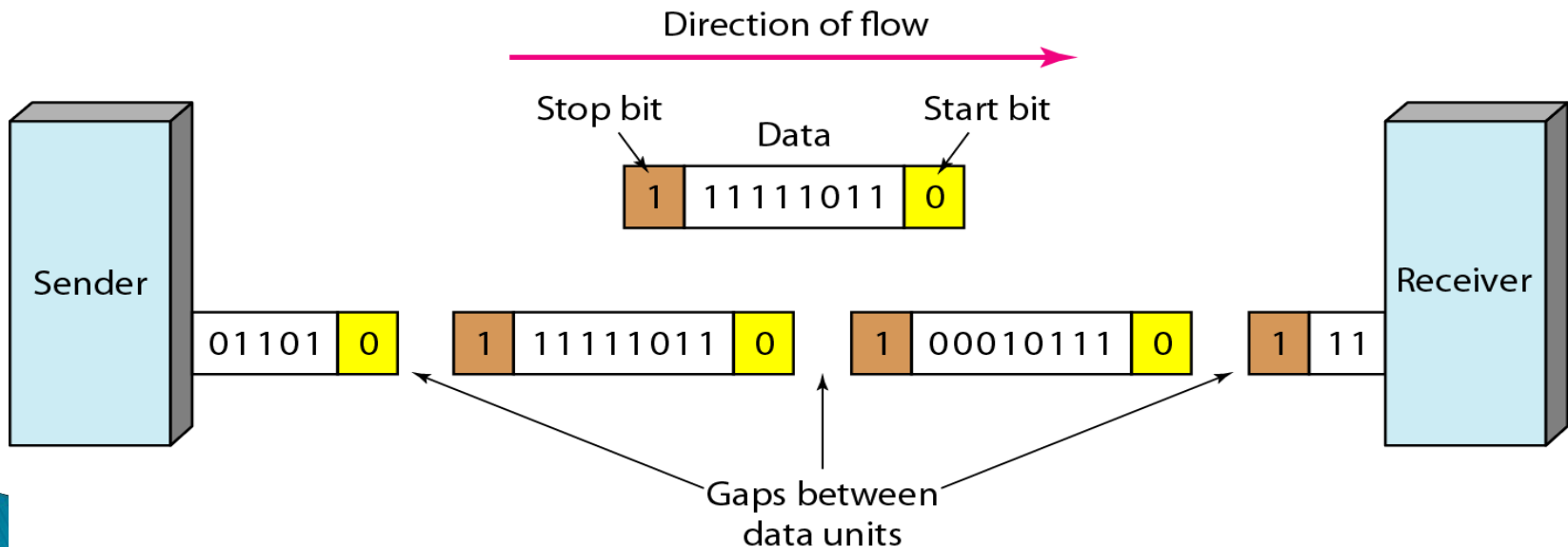
- their durations are the same

Asynchronous transmission

Slower Transmission due to additional bits

Cheap and effective

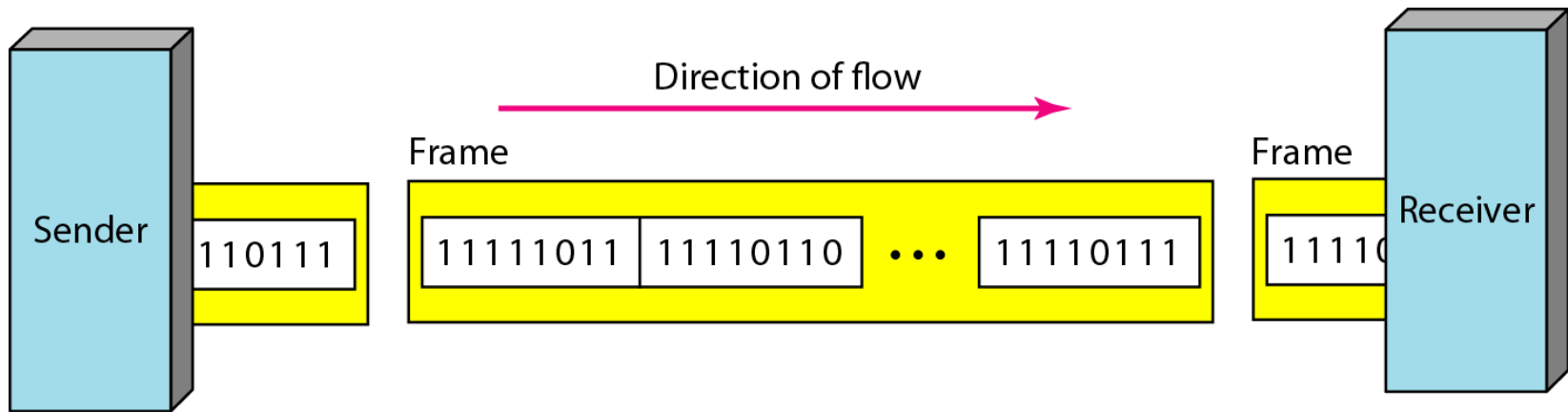
Used for low speed communication (between keyboard and computer)



Synchronous transmission

In synchronous transmission,

- we send bits one after another without start or stop bits or gaps
- It is the responsibility of the receiver to group the bits
- The bits are usually sent as bytes and many bytes are grouped in a frame
- A frame is identified with a start and an end byte and there are uneven gaps between frames



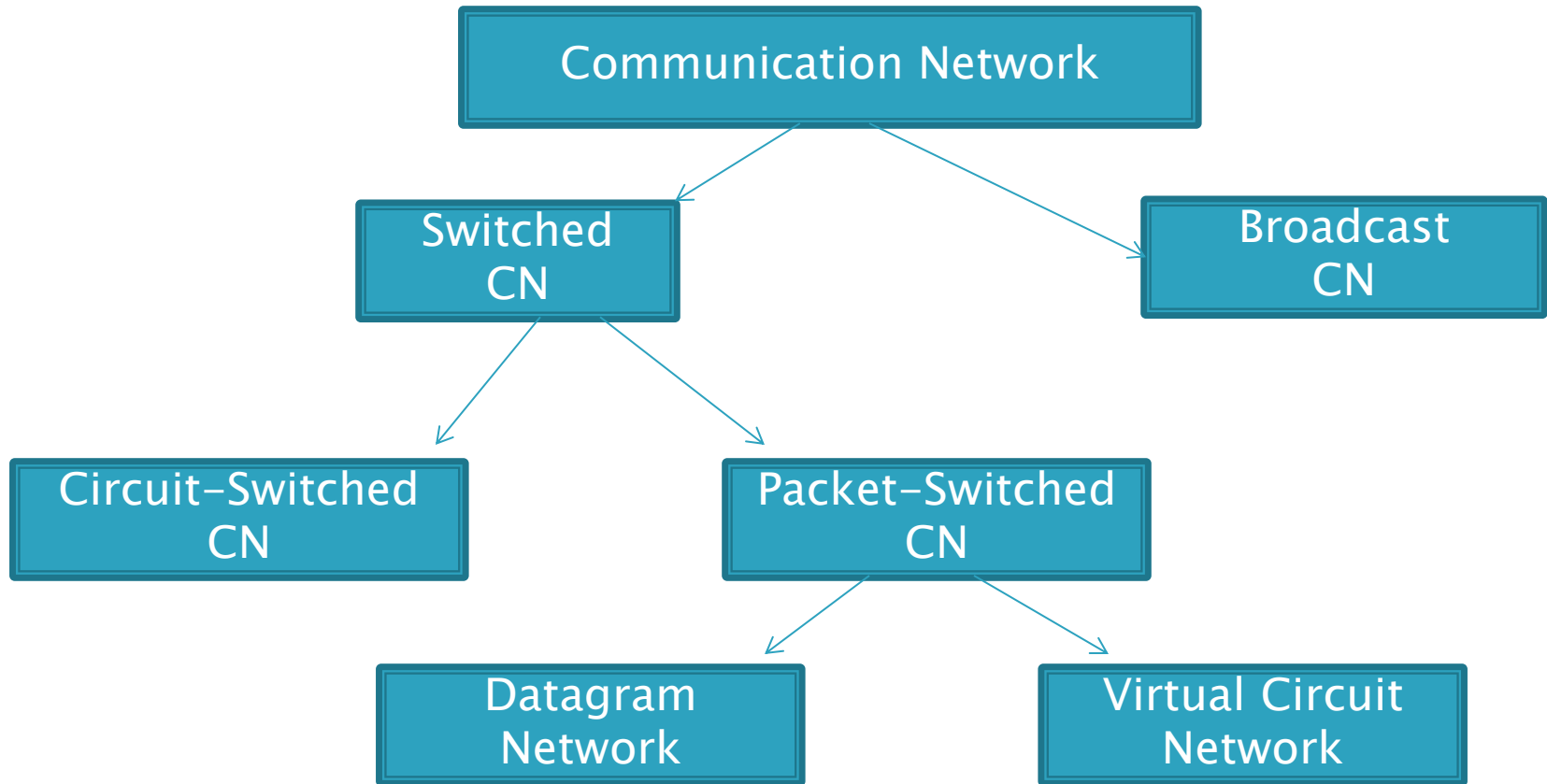
Isochronous

In isochronous transmission

- ▶ we cannot have uneven gaps between frames
- ▶ Transmission of bits is fixed with equal gaps
- ▶ Used in real-time audio and video where uneven delays are not acceptable
- ▶ Synchronization in entire stream of bits is needed

Taxonomy of Communication Networks

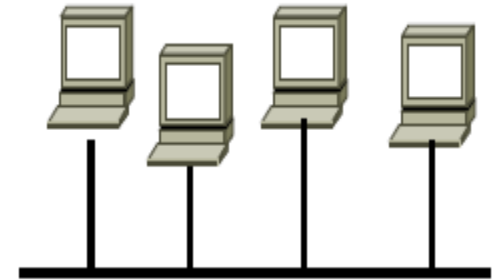
Communication Network (CN) can be classified based on the way in which the nodes exchange information



Broadcast Vs Switched Network

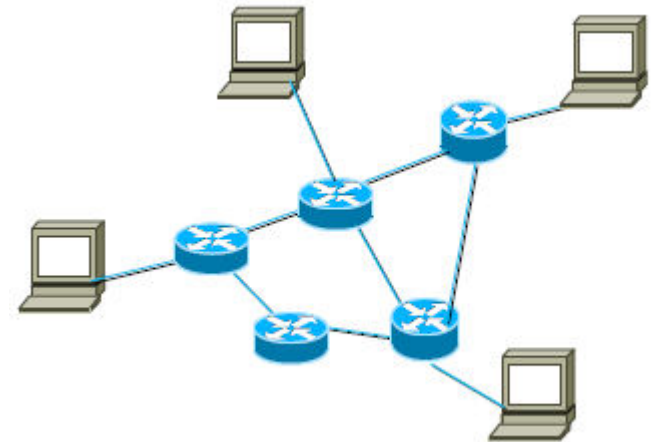
Broadcast networks

- Information transmitted by any node is received by every node in the network
- Ex:: Broadcast Ethernet, wireless LANs
- Need to coordinate the access to the shared medium
 - ❖ MAC



Switched networks

- Links are point-to-point
- Ex: WANs (Telephony Network, Internet)
- Routing becomes harder



Switched Network

- **Switched Network** is a series of interlinked nodes which are called switches
- In a switched network, some switching nodes are connected to the **end systems** (like computers, telephones), others are used only for **routing**

Switching: Methods

- Circuit switching
- Packet switching
 - ❖ Datagram approach
 - ❖ Virtual circuit switched approach

Circuit Switched Network

- Three phases in circuit switching
 - ❖ Establish
 - ❖ Transfer
 - ❖ Disconnect
- The telephone message is not broken
 - ❖ It is sent all together
- The message arrives in the same order as it was sent originally
- Electronic signals pass through many switches before a connection is established (**In modern circuit-switched networks**)

Circuit Switched Network

- During a call (**transfer phase**), switches can not be used by any other network traffic
 - ❖ Hence, the resources remain **dedicated** to the circuit during the entire transfer of data and the entire message follows the **same path**
- A circuit-switched network is excellent for data that needs a **constant link from end-to-end**, for example, real-time video

Packet Switched Network

- Packets are sent as soon as they are available
 - ❖ the message is broken into small data packets

Packet Switched network Approaches

- Datagram Network Approach
- Virtual Circuit Network Approach

Packet Switched Network

Datagram approach of packet switching

- **no need to set up a dedicated path in advance**
- It is up to routers to use store-and-forward transmission to send each packet on its way to the destination on its own
- Packets seek out the most efficient route to travel as circuits become available
 - not necessarily the shortest route
- **There is no fixed path**
 - Different packets can follow different paths
 - Packets may arrive out of order
- It places a tight upper limit on the size of packets
 - This ensures that no user can monopolize any transmission line for very long (e.g., many milliseconds)
 - Thus, **it can handle interactive traffic**

A comparison of circuit switched and Datagram packet switched networks

Item	Circuit switched	Packet switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
Time of possible congestion	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Charging	Per minute	Per packet

In Circuit and Packet Switched Network

- The trade-off is between guaranteed service and wasting resources versus not guaranteeing service and not wasting resources

Virtual Circuit Network

A cross between a circuit-switched network and a datagram network

➤ **As in a circuit-switched network**

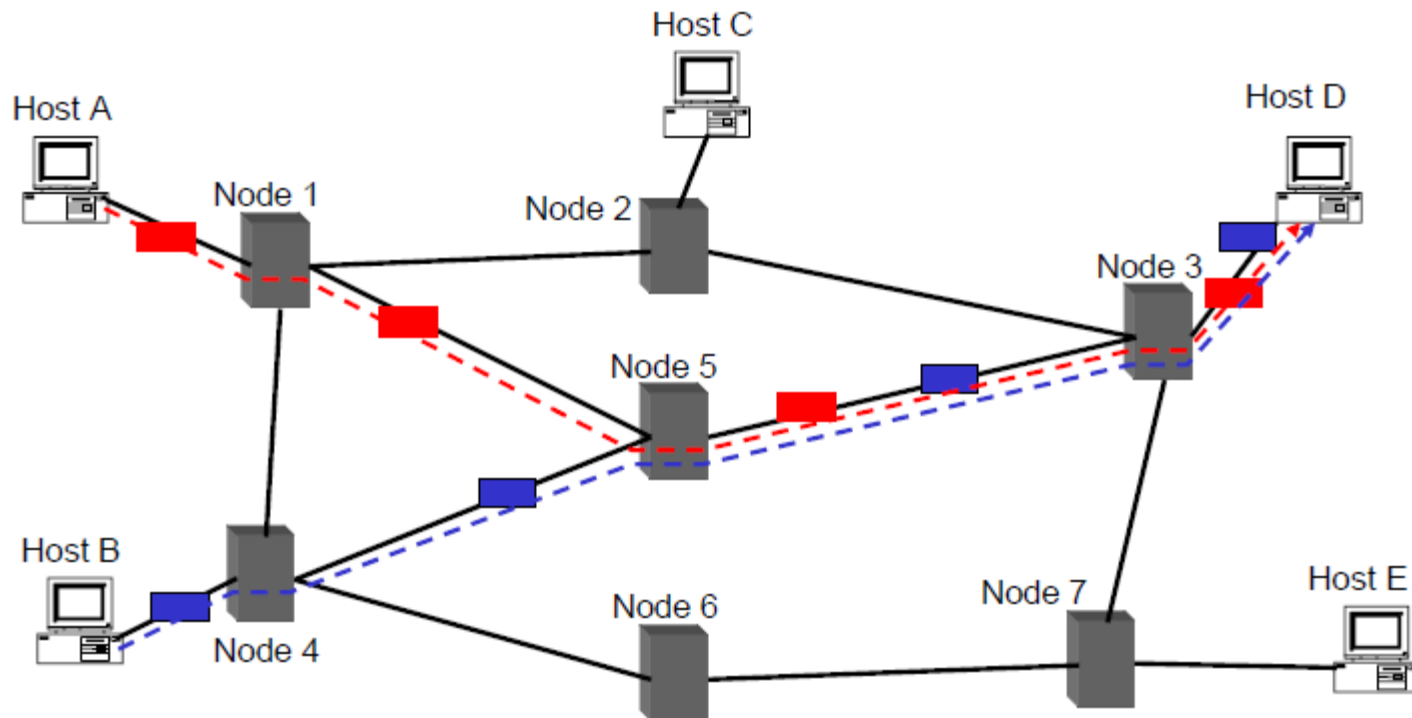
- ❖ It has setup and teardown phases in addition to the data transfer phase
- ❖ A virtual circuit is made before actual data is transmitted **but** it is different from circuit switching in a sense that
 - ❖ **in circuit switching** the **call accept** signal comes only from the final destination to the source
 - ❖ **while** in case of **virtual-packet switching** this **call accept** signal is transmitted between each adjacent intermediate node.
- ❖ all packets follow the same path established during the connection

Virtual Circuit Network

In virtual-circuit packet switching

- An initial setup phase is used to set up a route between the intermediate nodes for all the packets passed during the session between the two end nodes.
- In each intermediate node, an entry is registered in a table to indicate the route for the connection that has been set up.
- Thus, packets passed through this route, can have short headers, containing only a *virtual circuit identifier (VCI)*, and not their *destination*.
- This approach is **slower than Circuit Switching**, since **different virtual circuits** may **compete** over the same resources, and an initial setup phase is needed to initiate the circuit.

Virtual Circuits guarantees that all packets in a flow follow one path



Virtual Circuit Network

- Data are packetized and each packet carries an address in the header **as in datagram networks**
- But the address in the header has **local jurisdiction** which defines what should be the **next switch** and the **channel** on which the packet is being carried
- A **virtual-circuit** network (ATM and X.25)
 - normally implemented in the **data link layer**
 - No capacity guarantees, but guarantees no reordering of packets
- **circuit-switched** network
 - implemented in the **physical layer**
- A **datagram** network
 - Implemented in the **network layer**

INTERNET (in reality)

- is a datagram network
- BUT part of the Internet uses circuit-switching (Phone links) or virtual circuit (ATM)

IEEE Project 802

- As TCP/IP does not specify any protocol for data link and physical layer;
 - it accepts any protocol at these two layers that can provide services to network layer.
 - These two layers belong to networks (wired or wireless) that are using them.
- A LAN is computer network designed for a limited geographic area such as buildings or a campus.
- Most LANs are linked to a wide area network or Internet

IEEE Project 802

➤ In 1985, the Computer Society of the IEEE started a project, called **Project 802**, to set **standards** to enable **intercommunication** among **equipment** from a **variety of manufacturers**.

➤ Project 802 does not seek to replace any part of OSI or TCP/IP suit; it is a way of specifying functions of the **physical** layer and the **data link** layer of major LAN protocols

➤ **IEEE 802.3: Ethernet LAN**

➤ IEEE 802.4: Token bus

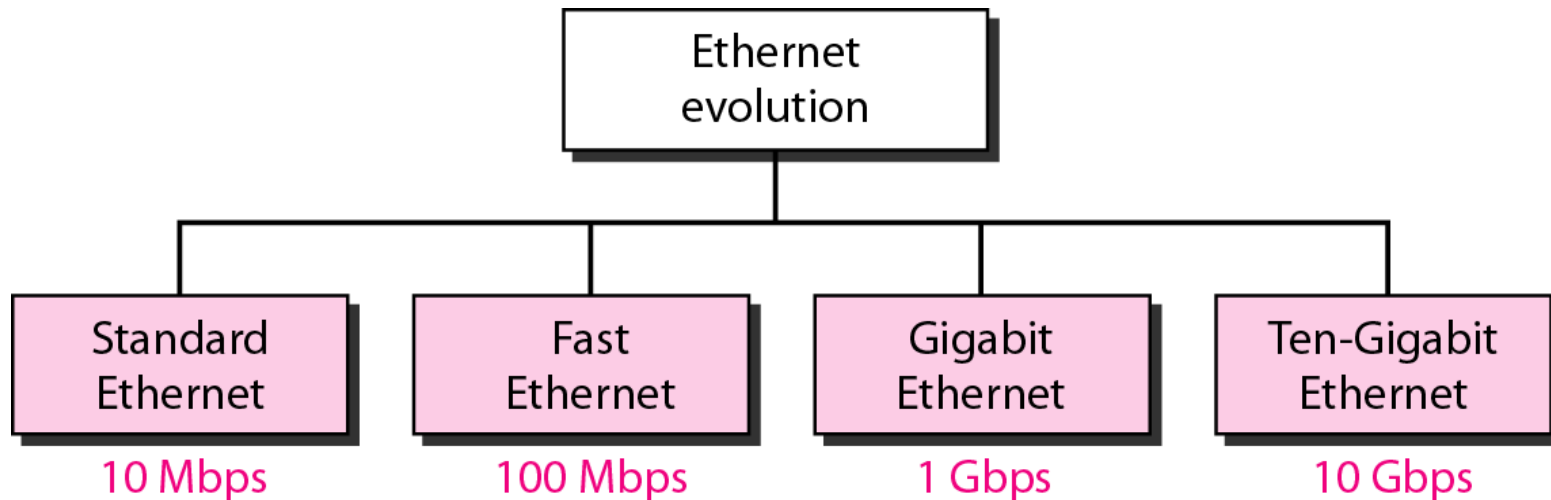
➤ IEEE 802.11: Wireless LAN (WLAN)

➤ IEEE 802.15: Wireless PAN (802.15.1 is for Bluetooth)

➤ IEEE 802.16: WiMAX

Standard Ethernet

The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC) in California. Since then, it has gone through four generations.



Standard Ethernet implementations

<i>Characteristics</i>	<i>10Base5</i>	<i>10Base2</i>	<i>10Base-T</i>	<i>10Base-F</i>
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

The 10-Mbps Standard Ethernet has gone through several changes before moving to the higher data rates.

These changes actually opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs.

Fast Ethernet

Fast Ethernet was designed to compete with LAN protocols such as FDDI (**Fiber Distributed Data Interface**) or Fiber Channel. IEEE created **Fast Ethernet** under the name **802.3u**. Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data **10 times** faster at a rate of **100 Mbps**.

Fast Ethernet implementations

<i>Characteristics</i>	<i>100Base-TX</i>	<i>100Base-FX</i>	<i>100Base-T4</i>
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T

Gigabit Ethernet

The need for an even higher data rate resulted in the design of the Gigabit Ethernet protocol (**1000 Mbps**). The IEEE committee calls the standard **802.3z**.

In the full-duplex mode of Gigabit Ethernet, there is no collision;

the maximum length of the cable is determined by the signal attenuation in the cable.

Gigabit Ethernet implementations

<i>Characteristics</i>	<i>1000Base-SX</i>	<i>1000Base-LX</i>	<i>1000Base-CX</i>	<i>1000Base-T</i>
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

Ten-Gigabit Ethernet implementations

<i>Characteristics</i>	<i>10GBase-S</i>	<i>10GBase-L</i>	<i>10GBase-E</i>
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-nm single mode
Maximum length	300 m	10 km	40 km

Example of an Ethernet address in hexadecimal notation

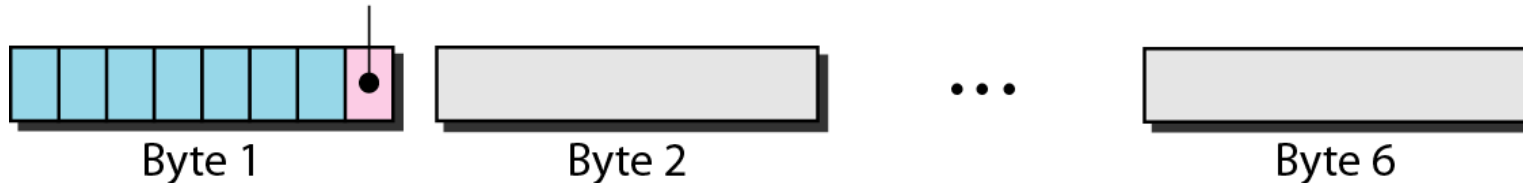
Each station on Ethernet network has its own Network Interface card (NIC) which provides the station with link layer address (6 bytes for Ethernet)

06 : 01 : 02 : 01 : 2C : 4B

6 bytes = 12 hex digits = 48 bits

Unicast and multicast addresses

Unicast: 0; multicast: 1



The **least significant** bit of the first byte defines the **type of address**. If the bit is **0**, the address is **unicast**; otherwise, it is **multicast**.

The **broadcast** destination address is a special case of the multicast address in which all bits are 1s.

The source address is always a unicast address

Example

Define the type of the following destination addresses:

- a. 4A:30:10:21:10:1A
- b. 47:20:1B:2E:08:EE
- c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, look at the second hexadecimal digit from the left.

If it is even, the address is unicast.

If it is odd, the address is multicast.

If all digits are F's, the address is broadcast.

- a. This is a unicast address because A in binary is 1010.
- b. This is a multicast address because 7 in binary is 0111.
- c. This is a broadcast address because all digits are F's.

Example

Show how the address **47:20:1B:2E:08:EE** is sent out on line.

Solution

The address is sent **left-to-right, byte by byte**;
for **each byte**, it is sent **right-to-left, bit by bit**;



Transmission in Ethernet is always broadcast (without regard the intention of unicast, multicast, broadcast)

Ethernet

Why Ethernet became so popular

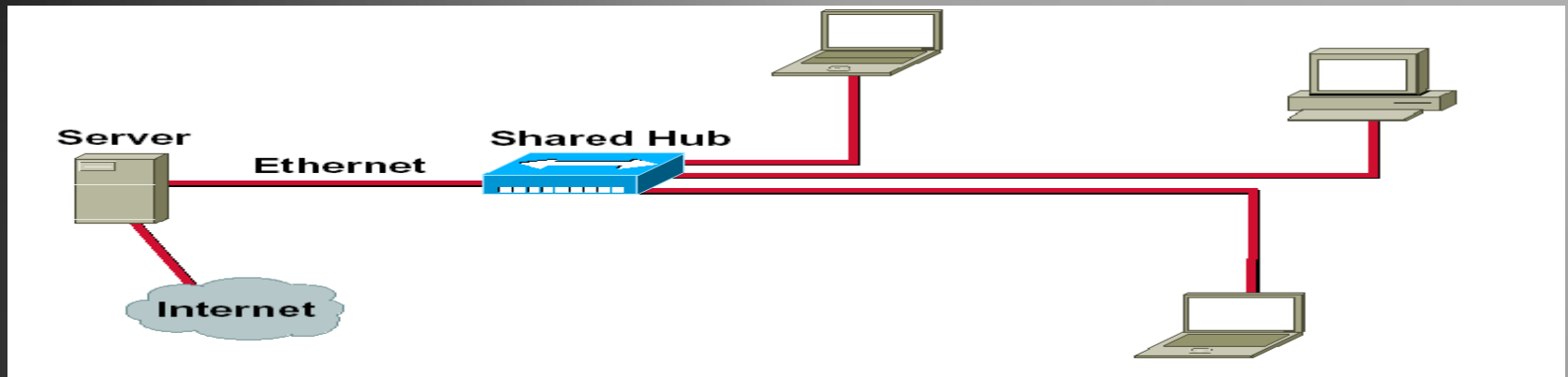
- Easy to understand, implement, manage, and maintain
- Low-cost network implementations
- Topological flexibility for network installation
- Successful interconnection and operation of products, regardless of manufacturer

Wireless LAN (WLAN)

- A wireless LAN uses wireless transmission medium
- WLAN provides wireless network communication over short distances
 - ❖ using radio or infrared signals instead of traditional network cabling like UTP
- Wireless LAN provides all the features and benefits of traditional LAN technologies such as Ethernet and Token Ring
 - ❖ but without the limitations of wires or cables

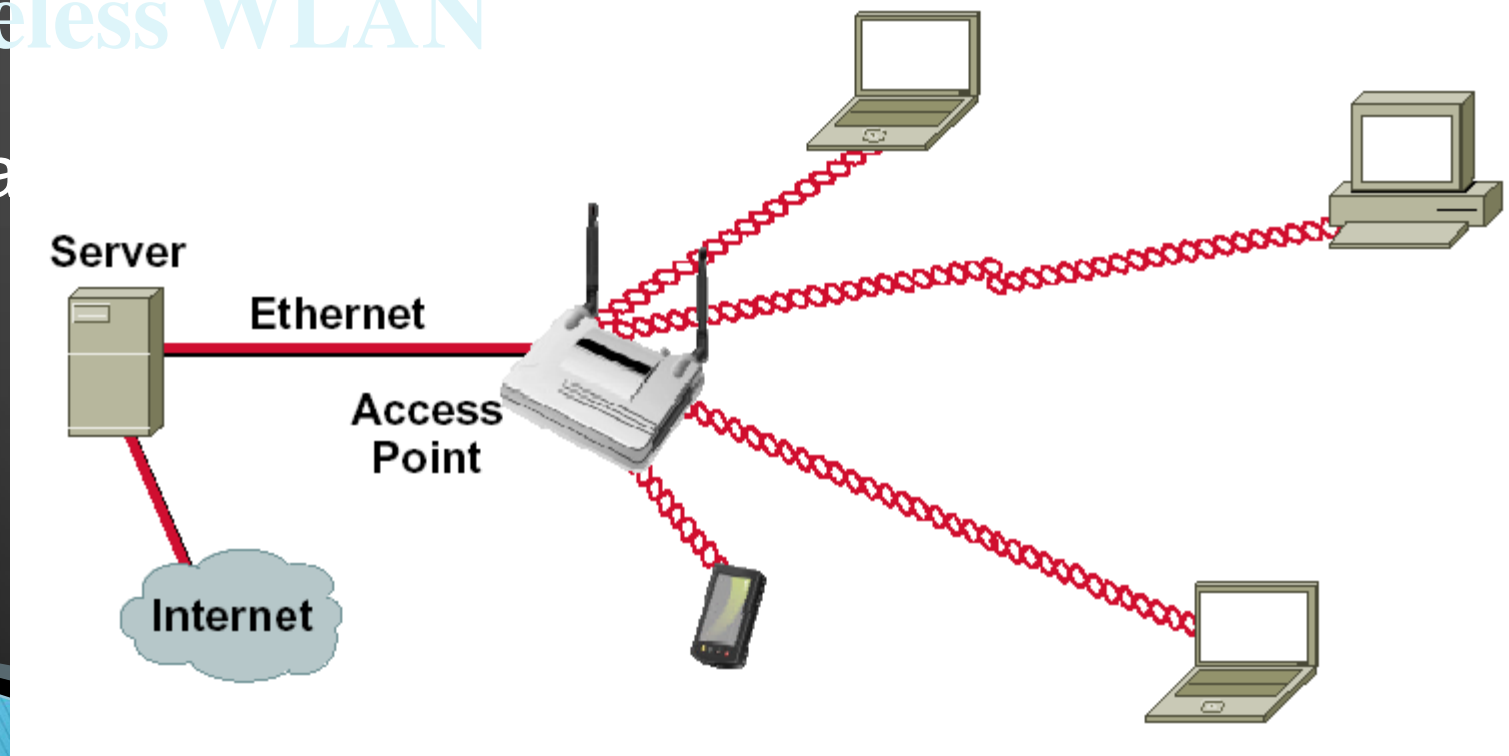
Popularity of wireless LANs has grown rapidly

A typical LAN



A Wireless WLAN

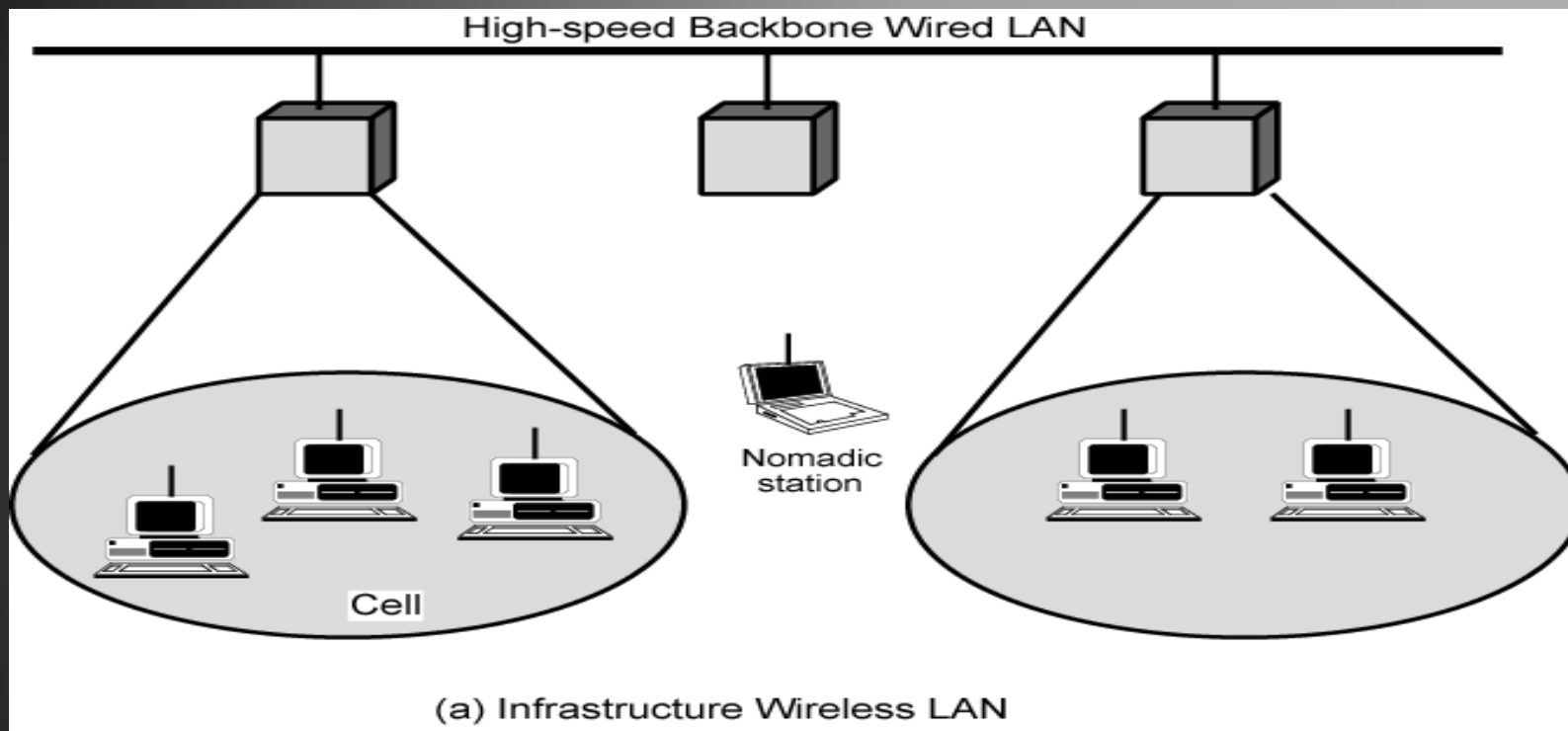
It's a



Wireless LAN

- A WLAN typically extends an existing wired LAN
- The *access point* (AP) is attached to the edge of the wired network to built a WLAN
 - a wireless *network adapter* enables clients to communicate with the AP
 - similar in function to a traditional Ethernet adapter
- WLANs use the **900 MHz**, **2.4 GHz** and **5-GHz** frequency bands.
- ISM (Industry, Scientific, Medical) license-free (unlicensed) frequency bands

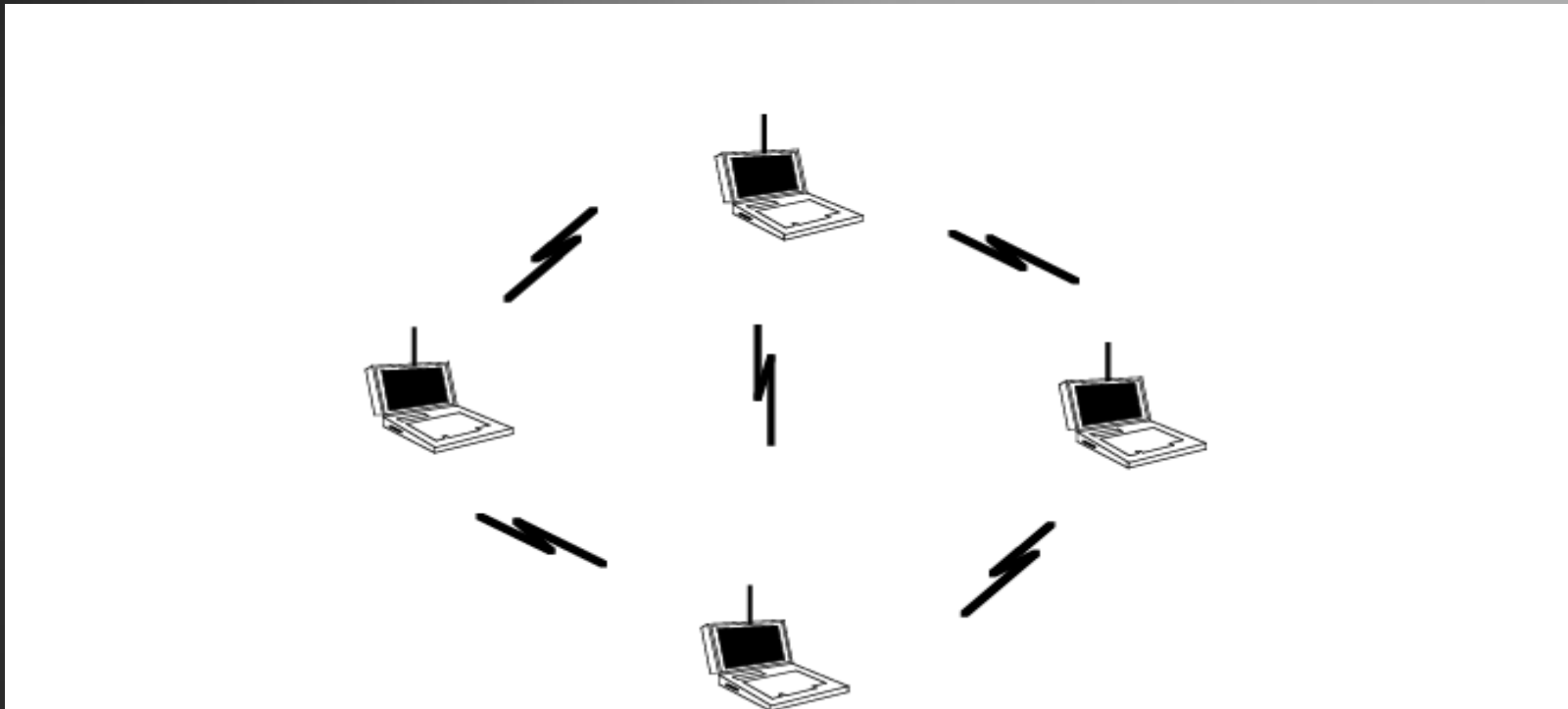
Infrastructure Wireless LAN



Wireless LAN forms a **stationary infrastructure** consisting of one or more cells with a control module for each cell

- Within a cell, there may be a number of stationary end systems.
- **Nomadic stations** can move from one cell to another

Ad Hoc LAN



- No infrastructure for an ad hoc network.
- A wireless network adapter is required to be installed
- a peer collection of stations within range of each other may dynamically configure themselves into a temporary network

WLAN Technology

according to transmission technique being used

Infrared (IR) LANs: Individual cell of IR LAN limited to single room

- IR light does not penetrate opaque walls
- Line of sight only

Spread spectrum LANs: Mostly operate in ISM (industrial, scientific, and medical) bands

- So no Federal Communications Commission (FCC) licensing is required in USA

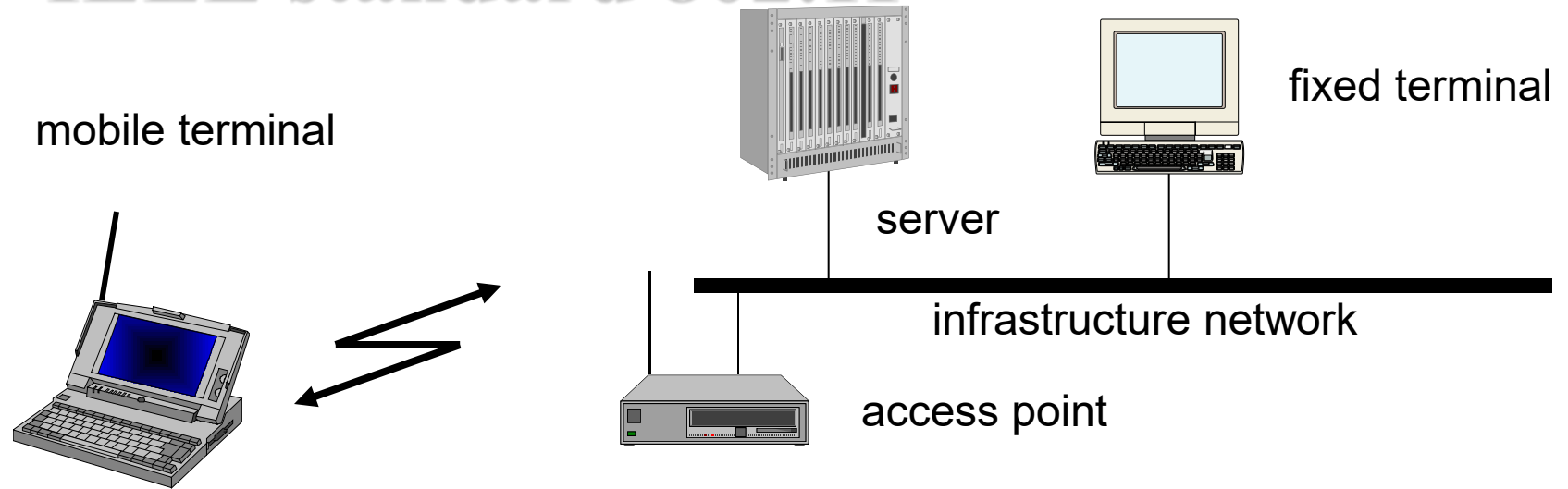
Narrowband microwave: Microwave frequencies but not use spread spectrum

- Some products operate at frequencies that require FCC licensing
- While others use one of the unlicensed ISM bands

IEEE 802.11

- In IEEE 802.11 std, the addressable unit is station (STA)
 - **Station (STA)**: a device that has the capability to use the 802.11 protocol
 - STA may be fixed, mobile or portable
- According to IEEE 802.11-2007 : **A STA** is *any device that contains an IEEE 802.11-conformant media access control (MAC) and physical layer (PHY) interface to the wireless medium (WM).*
- One requirement of IEEE 802.11 is to handle *mobile as well as portable STAs*
- **A portable STA** is one that is moved from location to location, but that is only used while at a fixed location.
- **Mobile STAs** actually access the LAN while in motion.

IEEE standard 802.11



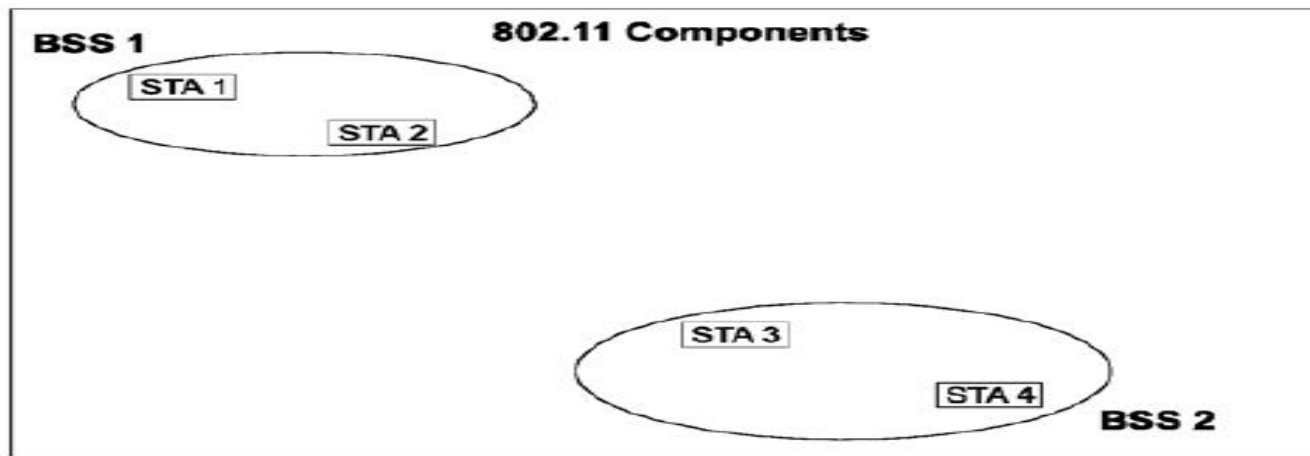
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TCP
IP
LLC
802.11 MAC
802.11 PHY

LLC	
802.11 MAC	802.3 MAC
802.11 PHY	802.3 PHY

application
TCP
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802.3 PHY

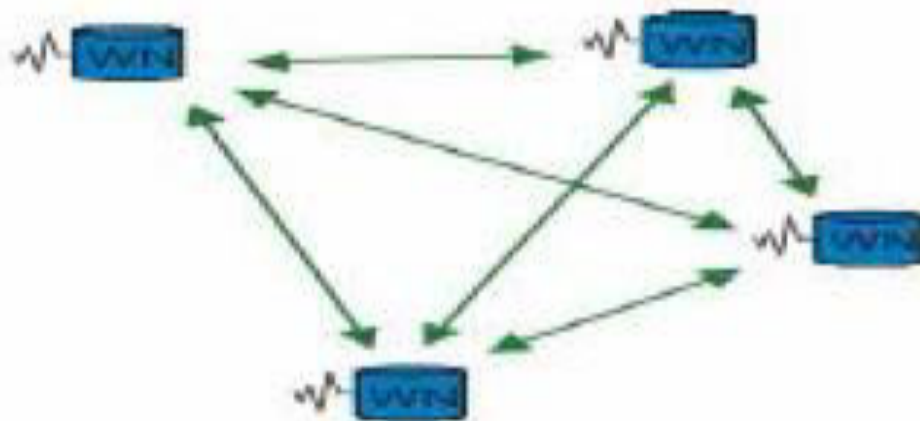
Components of IEEE 802.11 architecture

- **The basic service set (BSS)** is the basic building block of an IEEE 802.11 LAN
- Each of BSS1 and BSS2 has two STAs that are members of their BSS
- Think of the ovals as coverage area of a BSS within which the member STAs may remain in communication
- This area is called the **Basic Service Area (BSA)**.
 - If a STA moves out of its BSA, it can no longer directly communicate with other STAs present in the BSA



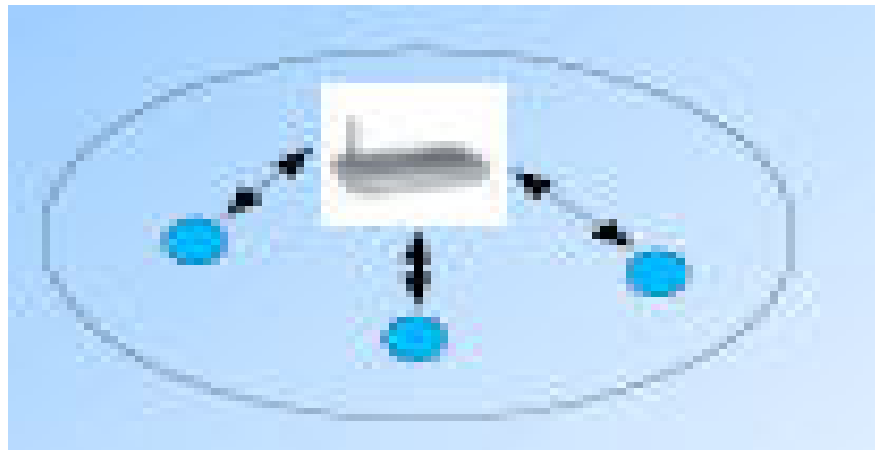
Components of IEEE 802.11 architecture

- **Two type of BSS:** Independent and Infrastructure BSS
- Every BSS has an id called the BSSID, it is the MAC address of the access point servicing the BSS
- **Independent BSS (IBSS)** is simply comprised of one or more Stations which communicate directly with each other (ad-hoc network)
 - They contain no Access Points
 - They can not connect to any other basic service set



Components of IEEE 802.11 architecture

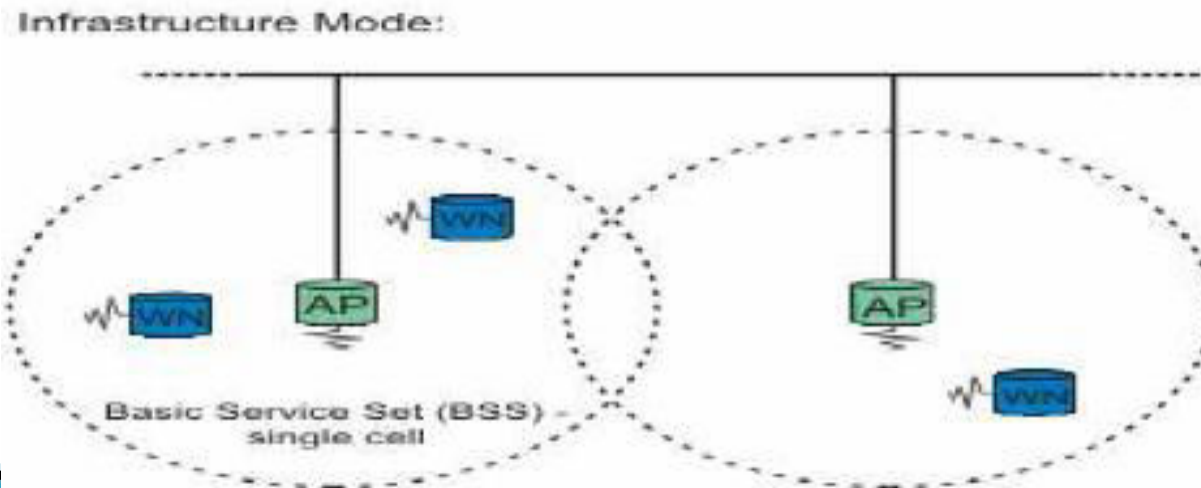
- In Infrastructure BSS,
 - STAs communicate with each other through Access Points
 - STAs can communicate with other stations not in the same basic service set through Access Points



Components of IEEE 802.11 architecture

- **An Extended Service Set (ESS)** is a set of connected BSS
- Access Points in an extended service set are connected by a distribution system
- Each ESS has an ID called the SSID

An ESS is the union of the infrastructure BSSs with the same SSID connected by a DS



Components of IEEE 802.11 architecture

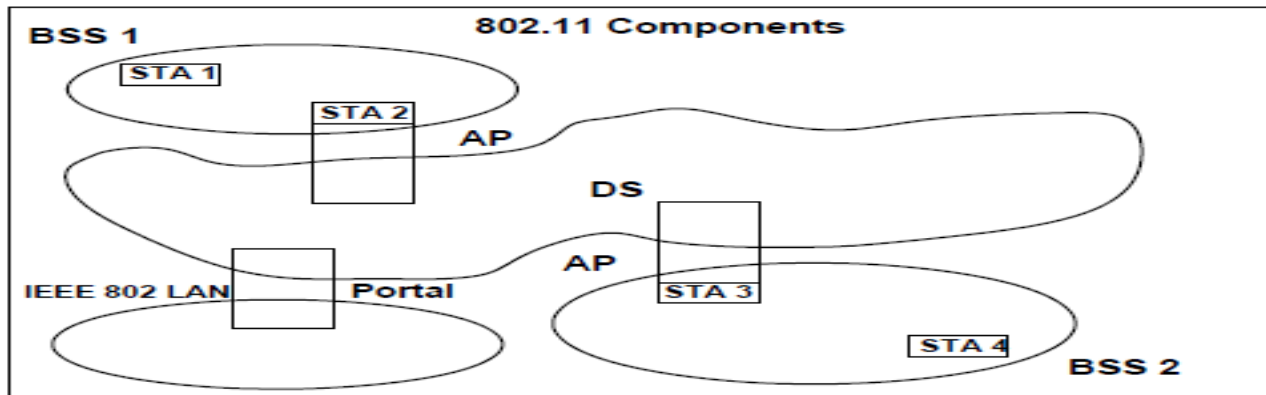
A Distribution system (DS)

- connects Access Points in an extended service set
- is usually a wired LAN but can be a wireless LAN
- Is the architectural component used to interconnect infrastructure BSSs (IEEE 802.11 2012)

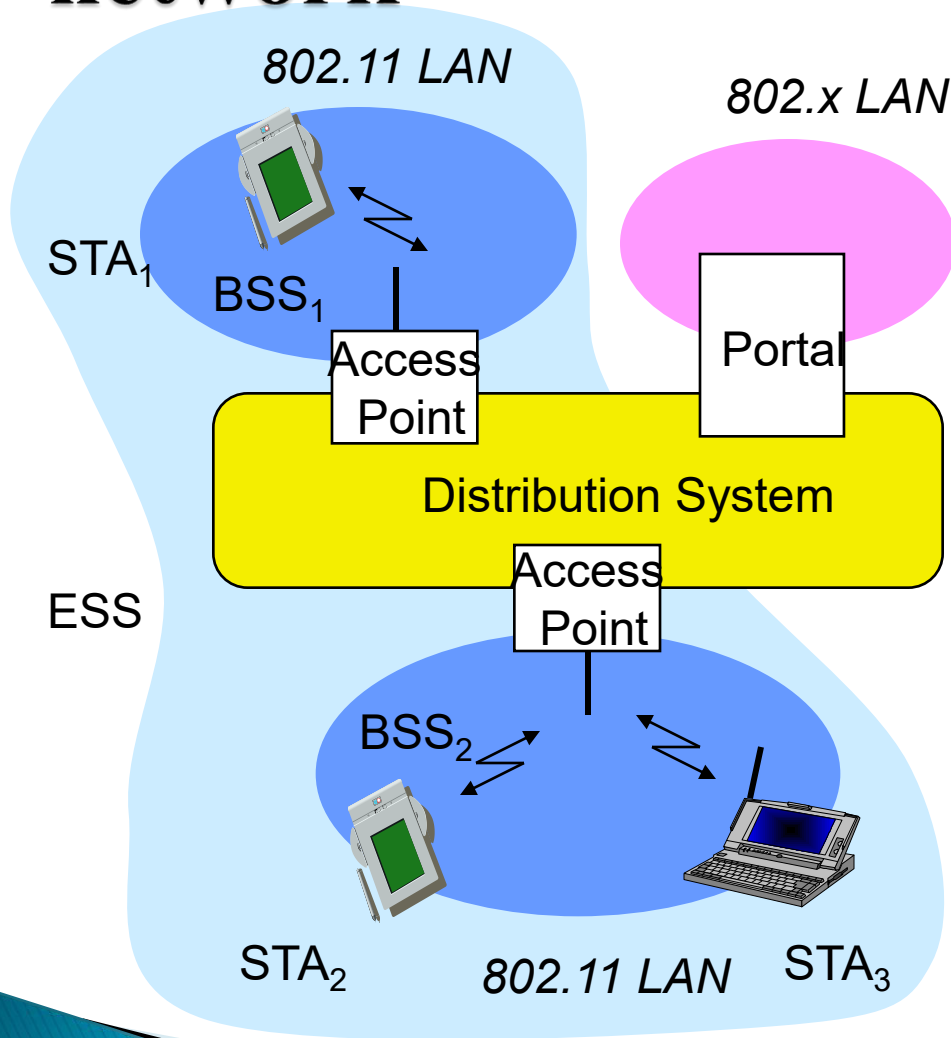
Components of IEEE 802.11 architecture

Portal bridge to other (wired) networks

- A portal is the logical point at which MSDUs (MAC service data units) from an integrated non-IEEE-802.11 LAN enter the IEEE 802.11 DS
 - In other words, All data from non-IEEE-802.11 LANs enter the IEEE 802.11 architecture via a portal
- It is possible for one device to offer both the functions of an AP and a portal (IEEE 802.11 2012)
- **The portal logic is implemented in a device such as bridge or router, that is part of the wired LAN and that is attached to the DS**

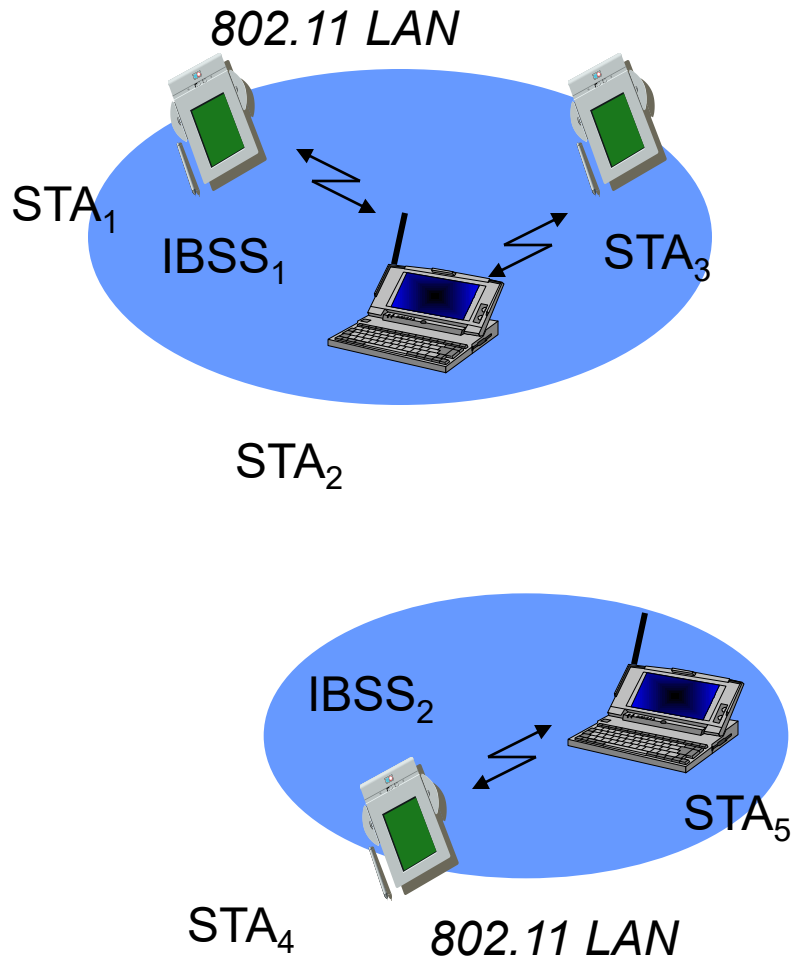


802.11 Architecture: Infrastructure network



- **Station (STA)**
 - terminal with access mechanisms to the wireless medium and radio contact to the access point
- **Basic Service Set (BSS)**
 - group of stations using the same radio frequency
- **Access Point**
 - station integrated into the wireless LAN and the distribution system
- **Portal**
 - bridge to other (wired) networks
- **Distribution System**
 - interconnection network to form one logical network (EES: Extended Service Set) based on several BSSs

802.11 Architecture: ad-hoc network



Direct communication within a limited range

- **Station (STA):**
terminal with access mechanisms to the wireless medium
- **Independent Basic Service Set (IBSS):**
group of stations using the same radio frequency without a controlling access point